



Society of Petroleum Engineers

SPE-228935-MS

Autonomous Directional Drilling and Geosteering for Improved Real-Time Decisions Using UDAR in Maximum Reservoir Contact Wells: Offshore Abu Dhabi Case Study

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/228935-MS](https://doi.org/10.2118/228935-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Drilling Maximum Reservoir Contact (MRC) wells, which are essential for maximizing reservoir exposure and enhancing production. These Maximizing Reservoir Contact (MRC) wells play a vital role in the efficient development of mature fields. These complex operations demand precise wellbore trajectory control during the landing, lateral drilling, and geosteering phases to ensure optimal reservoir exposure. In high-density drilling environments, such as offshore Abu Dhabi, the risk of collision with existing wells necessitates robust real-time survey management and anti-collision strategies. Traditionally, trajectory control relied heavily on the expertise of directional drillers, who manually adjusted bottomhole assembly (BHA) configurations and drilling parameters. However, this human-dependent approach often resulted in trajectory deviations and inconsistent well performance. This paper presents a novel, autonomous drilling methodology that integrates real-time directional drilling control with advanced geosteering software. The system autonomously adjusts the downhole Rotary Steerable System (RSS) to maintain the wellpath within the target zone, significantly reducing reliance on human intervention. Integration with geosteering software accelerates decision-making, reducing response time from approximately 15 minutes to less than 5 minutes. Moreover, minimizing RSS setting changes results in a smoother wellbore, improved BHA compatibility for production, and a reduction in Invisible Lost Time (ILT). The case study focuses on the successful application of this integrated system in a carbonate reservoir offshore Abu Dhabi. The well was drilled from a 7in whipstock sidetrack, landing, and drilling a 6in lateral section within a 45-ft-thick reservoir characterized by water encroachment risks, structural complexity, and geophysical uncertainties. The planned trajectory required up to 3.55°/100 ft dogleg severity during landing and less than 2°/100 ft in the lateral section. Offset well analysis highlighted the need for Ultra-Deep Resistivity (UDR) and density imaging to resolve subsurface uncertainties. The drilling BHA featured a 4.75in push-the-bit RSS equipped with UDR sensors positioned 65 ft behind the bit, along with real-time monitoring of inclination, azimuth, and drilling dynamics Weight On Bit (WOB) and Torque On Bit (TOB).

The lateral section was successfully drilled and geosteered over 7051 ft with 100% reservoir contact, guided entirely by the automated downhole directional drilling and geosteering platform, helped in reducing the Geosteering decisions and downlinking the RSS by more 60% compared to the offset well in same reservoir. This significant reduction facilitated smoother well trajectory and improved the rate of penetration (ROP) by 48%, reduced the frequency of downlinks to the RSS, and enabled faster, more consistent Geosteering decisions. This case study demonstrates the effectiveness of closed-loop downhole directional drilling automation in enhancing drilling efficiency and well placement accuracy, paving the way for broader application in similar reservoirs across UAE fields.

Introduction

In Abu Dhabi Offshore there is a field located in the central part of the Arabian Gulf at about 135 km NW of Abu Dhabi town. This field has Carbonate sedimentology and is a gentle slightly asymmetric domal anticline structure. The Field was discovered in January 1958. At present, the main crude production is from Arab reservoirs.

Formation A and B reservoirs are from the Upper Jurassic age and represent a large-scale southwest prograding carbonate ramp system. Typical facies succession consists of basal deeper water outer and mid ramp limestone (wackestone facies). Lateral variation of subsidence rate across the field likely reflects the impact of halokinesis. The lateral variability of depositional facies in this prograding system is further enhanced by superimposed diagenetic processes.

The log porosity ranges from 10 to 15 P.u and the avg permeability of around 8 mD in formation A and the log porosity ranges from 8 to 12 Pu and the avg permeability of around 2 mD in formation B.

The development of mature oil fields increasingly relies on drilling Maximum Reservoir Contact (MRC) wells, which are essential for maximizing reservoir exposure and enhancing production. Executing these complex wells demands precise trajectory control throughout critical stages, including well landing, horizontal drilling, and geosteering. One of the persistent challenges in such environments is the elevated risk of collisions with existing wellbores, necessitating the deployment of advanced, real-time anti-collision systems and rigorous survey management protocols. Historically, directional drillers (DDs) have managed wellbore trajectory by leveraging their field experience and understanding of Bottomhole Assembly (BHA) behavior within specific formations. They would make manual adjustments to drilling parameters and reconfigure BHAs as conditions evolved. While this hands-on approach has long been the norm, it is prone to human error and often results in unintended deviations from the planned well path, contributing to variable well performance. To address these limitations, this paper presents a next-generation solution: the integration of autonomous drilling technology with real-time geosteering automation. This approach removes much of the manual decision-making traditionally required to control the Rotary Steerable System (RSS). By relying on high-speed, precision software that continuously monitors and adjusts the well trajectory in real time, the system ensures that drilling remains accurately aligned with the reservoir target. The synergy between automated drilling and geosteering not only enhances trajectory accuracy but also reduces the need for frequent intervention and BHA reconfiguration. As a result, the system contributes to a smoother well profile, decreased Invisible Lost Time (ILT), and greater operational efficiency. Ultimately, this integrated automation framework represents a step-change in how complex wells are drilled—enabling consistent, high-performance outcomes that were previously difficult to achieve using conventional methods.

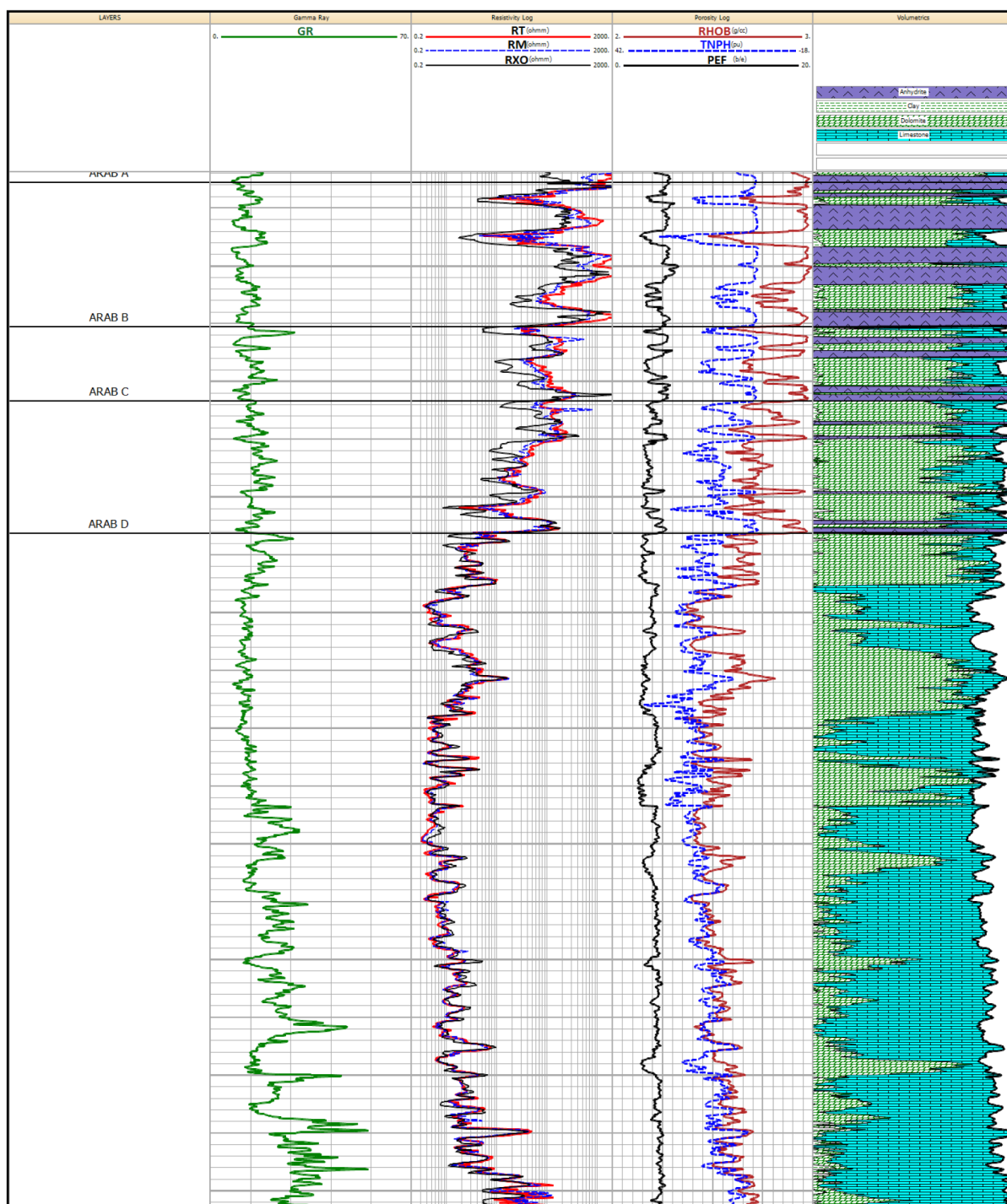


Figure 1—Field A Arab reservoir typical lithology, petrophysics, and sequence stratigraphy, 1:500-ft scale.

Solutions Overview

The autonomous drilling system is engineered to control the Bottomhole Assembly (BHA) with minimal manual input, freeing up engineers and drillers to focus on strategic and supervisory tasks. The platform leverages real-time data from advanced downhole sensors to monitor drilling conditions and execute decisions autonomously. Embedded anti-collision detection ensures safe operations in congested subsurface environments by continuously scanning for existing wells and geological obstructions. At the heart of this system lies an intelligent Rotary Steerable System (RSS), built using state-of-the-art technologies to deliver

accurate trajectory control and perform complex downhole computations. This RSS is capable of steering at speeds up to 400 rpm, with toolface measurements captured 1000 times per sec, allowing for high-resolution steering feedback. The system functions in concert with a suite of automated advisory algorithms that combine physical modeling, a digital twin of the BHA, and real-time data analytics enhanced by machine learning. The platform provides:

- **Steering Optimization** – Real-time guidance to predict and adjust the well trajectory, navigate geological hazards, and maintain the wellbore within optimal target zones.
- **Vibration Management** – Automated recommendations for optimal rotary speeds, helping to mitigate downhole vibration while improving directional response and BHA integrity.

Parallel to drilling automation, an integrated geosteering engine processes formation data in real time using resistivity imaging and other measurements. It identifies geological features, such as conductive layers, and detects changes in formation properties, allowing the drill bit to be steered precisely within the reservoir. The software dynamically updates the expected well trajectory and relays adjustments to the autonomous RSS to maintain target alignment and optimize reservoir exposure.

The system also employs a leading-edge ultra-deep resistivity (UDR) technology that can visualize formation boundaries and fluid contacts up to 225 ft (68 m) away from the wellbore. This service includes the industry's only true 3D inversion capability, offering highly accurate images of subsurface structures. In both offshore and aging onshore fields, this technology supports more informed development planning, reduces uncertainty, and avoids unnecessary sidetracks and pilot wells.

Core applications include:

- **Geostopping** – Enables accurate casing-point placement by identifying critical reservoir transitions early, reducing the need for pilot holes and preventing drilling into high-risk zones such as overpressured formations.
- **Geosteering** – Maximizes well productivity by keeping the borehole within the reservoir's most productive zones, while minimizing the risk of formation exits or inefficient drilling in low-value rock.
- **Geomapping** – Builds a detailed 3D model of the formation, identifying reservoir boundaries and bypassed zones. This insight supports reserve estimation and guides future well placement, especially in complex or mature reservoirs.

Altogether, this autonomous directional drilling solution—when combined with advanced geosteering and real-time formation evaluation—delivers a comprehensive system for efficient and intelligent well construction. By integrating high-frequency data collection, multi-disciplinary analytics, and automation, the platform enables rapid decision-making, improved trajectory control, and repeatable, high-quality well delivery in even the most challenging geological environments

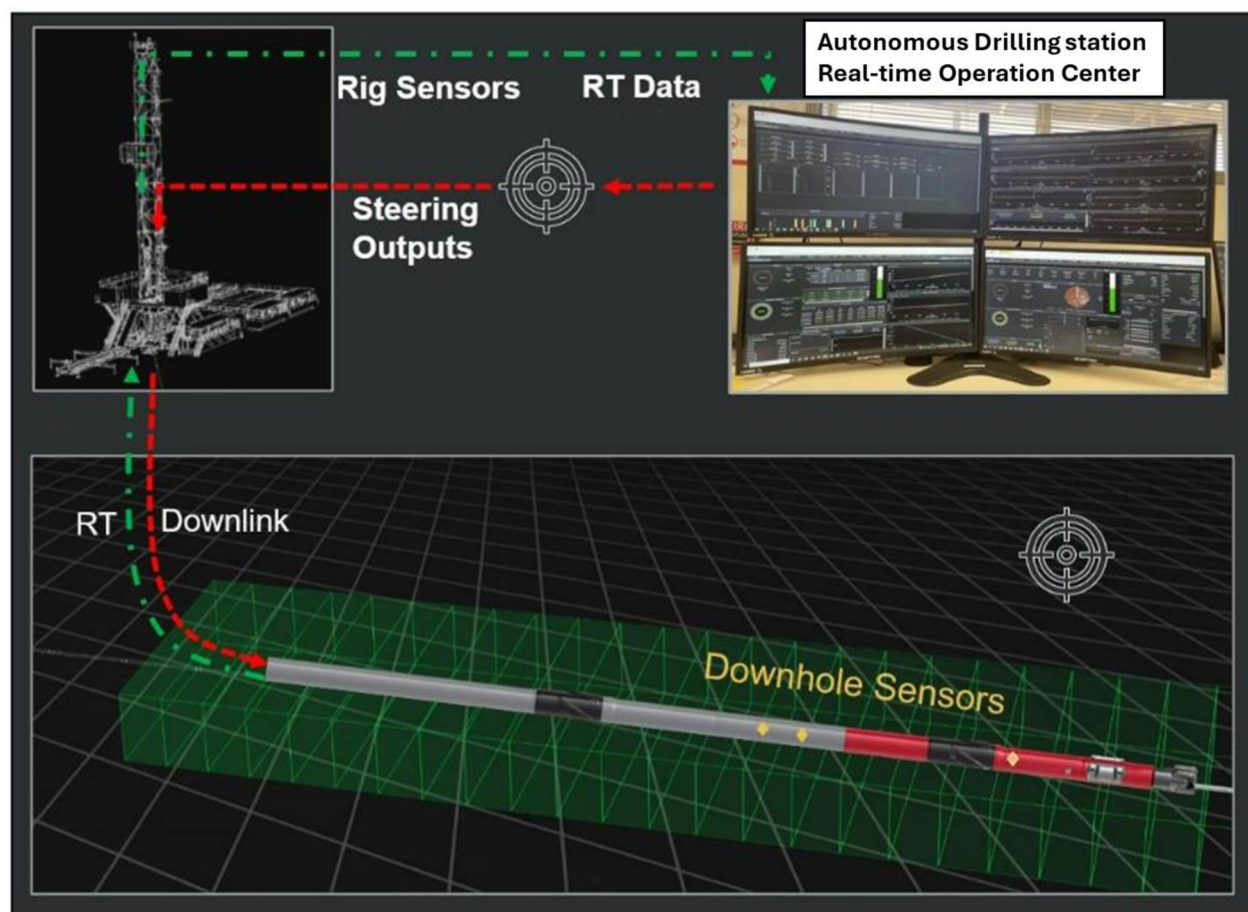


Figure 2—Autonomous Drilling Platform

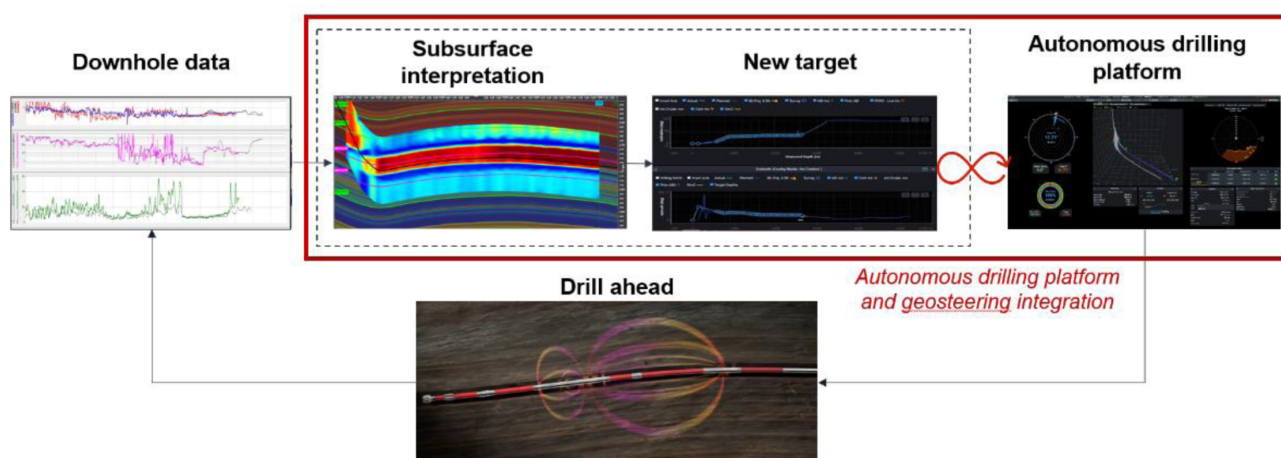


Figure 3—Workflow for Automated Well Placement

Planning phase

The automated downhole directional drilling and Geosteering workflow were planned to be implemented in a dual lateral of an oil producer well offshore Abu Dhabi. The 6in lower lateral drain will be drilled using conventional methods, while the 6in upper lateral drain will be drilled deploying the automated downhole directional drilling and Geosteering workflow. The well trajectory is depicted in Figure 3a below.

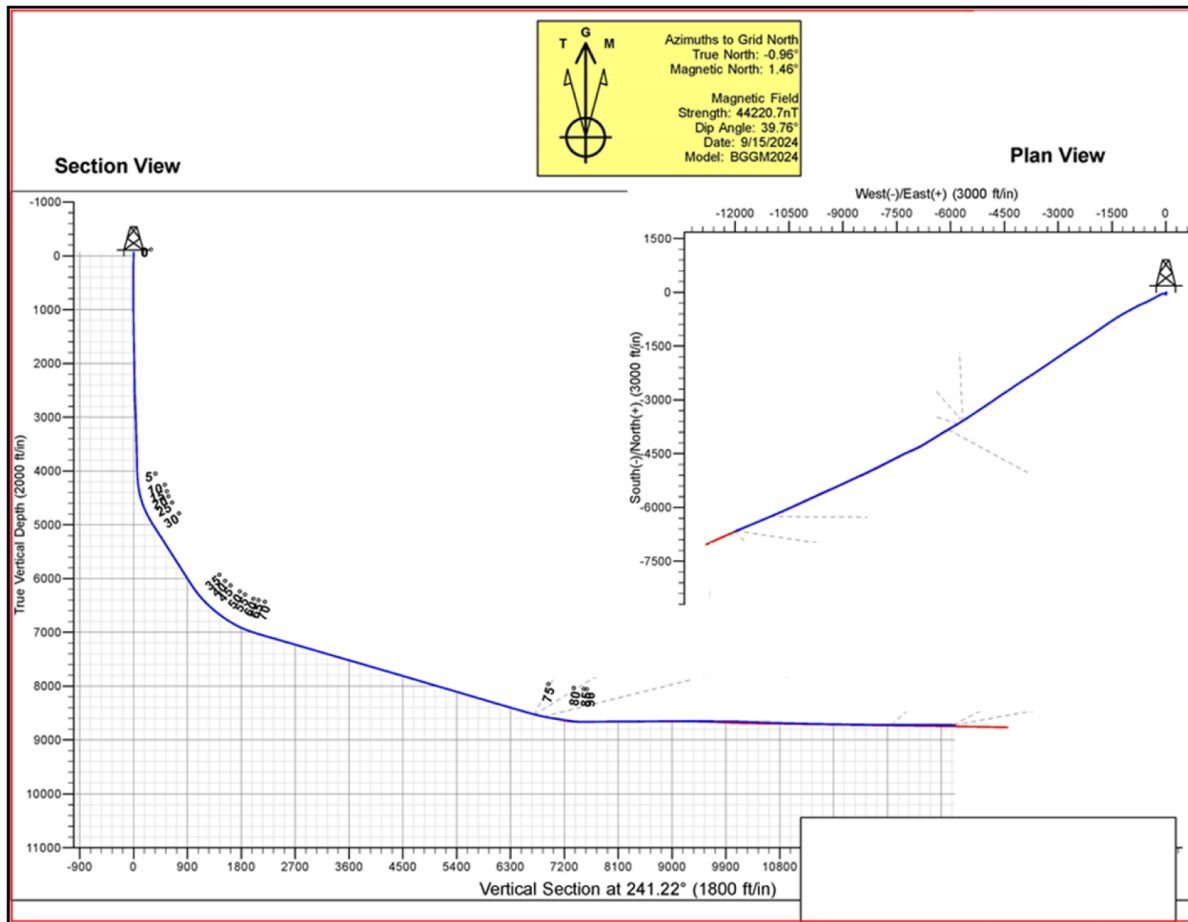


Figure 3a—Well Plan (Upper Lateral Drain)

The upper lateral drain is designed to target a carbonate reservoir with a thickness of 65 ft true vertical depth (TVD) between high water saturation zones, as illustrated in Figure 4 below:

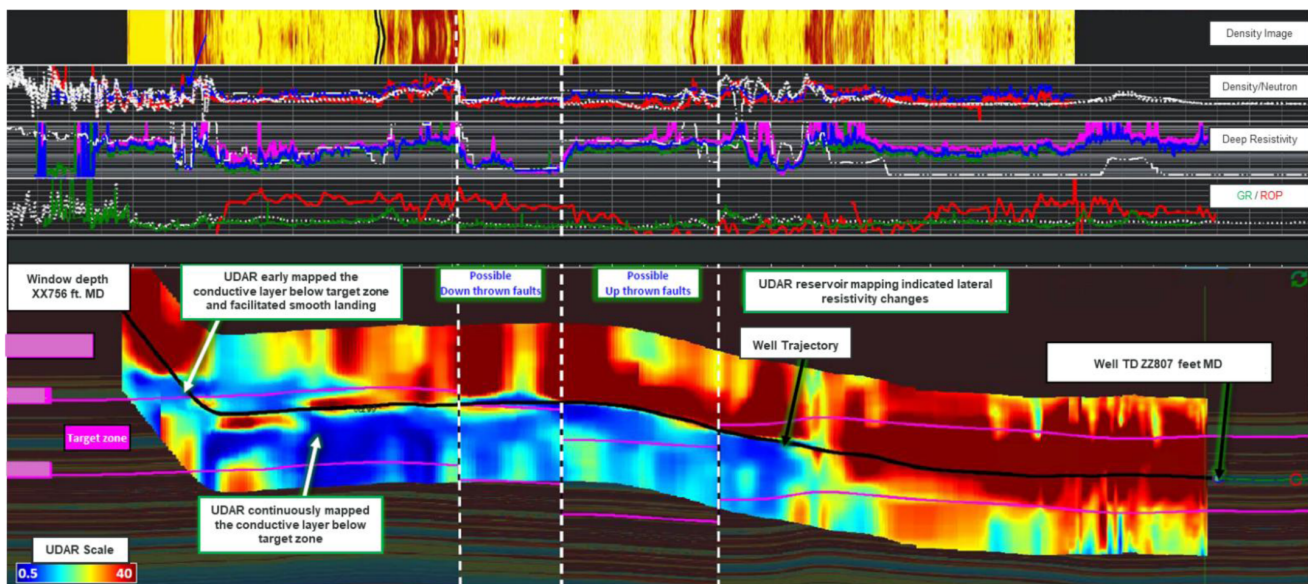


Figure 4—Upper Lateral Geosteering cross section

To facilitate the implementation of an The automated downhole directional drilling and Geosteering workflow, the Bottom Hole Assembly (BHA) is equipped with a push-the-bit rotary steerable system, Measurement While Drilling (MWD), and Logging While Drilling (LWD) tools, including Gamma Ray (GR), Resistivity (Res), Neutron (Neu), and Density (Den) sensors, as depicted in Figure 5 below. The push-the-bit rotary steerable system permits precise control over the drilling direction, ensuring accurate wellbore placement. The MWD provides real-time data on the well's trajectory and formation properties, while the LWD suite offers comprehensive geological information. Combined, these technologies enable efficient and effective decision-making during the drilling process, optimizing well placement and enhancing overall productivity.

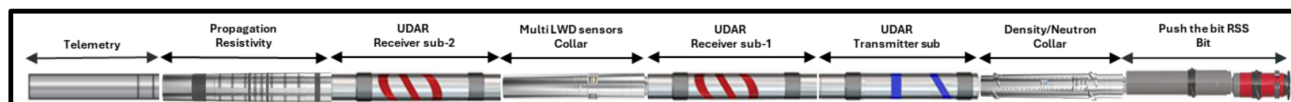


Figure 5—BHA Configuration

BHA Components:

- Push the bit – intelligent Rotary Steerable System
- Azimuthal Density / Neutron-Porosity
- Ultra Deep Resistivity – reservoir mapping
- Near bit Gamma Ray – conventional Gamma Ray
- Directional Module
- Downhole drilling dynamics
- Multi-frequency Propagation Resistivity

Sensor-to-Bit Measurement:

- ABI to bit 6.86 ft.
- At bit GR 14.31 ft.
- Direction to bit 17.52 ft.
- Density to bit 43.82 ft
- Porosity to the bit 52.95 ft.
- UDAR to bit 65.87 ft.
- Resistivity to the bit 153.38 ft.

To measure the effectiveness of the proposed solutions, the performance targets were set as follows:

- a. Reduce frequency of downlink by -10%
- b. Improve on-bottom rate of penetration (ROP) by $+15\%$
- c. Reduce number of geosteering decisions by -17%
- d. Assurance of real-time inclination & azimuth trend all over the run by 80%

Results

Drilling the upper lateral drain was challenging, where control of on-bottom ROP was experienced due to losses, high torque and vibration while switch drilling with sea water and presence of junk in hole which caused ring out and undergauge of drill bits. [Table-1](#) summarized the drilling event with RSS BHA until reached target depth.

Table 1—Drilling Activity

Depth	Activity
XX,243'	- Started drilling and steering 6in upper drain - Applied parameters: 150 rpm, 10-20 Klb WOB, 295 gpm
XY,800'	- Losses increased to 70-80 bbl/hr. - Continue drilling with sea water MW:9 ppg.
YY,951'	- Stop drilling, pumped 50bbls Lo Vis followed by 50bbls Hi Vis tandem pill twice and circulate hole clean. Torque reached 20 Klb, WOB reached max 20 lb and ROP avg dropped to 15-20 ft/hr. - Perform short wiper trip backream 10 stands, circulate each 5 stands bottom up for hole cleaning. - RIH to bottom on elevator, hole in good condition. - At bottom had very weak MWD signal, drill 8 ft blind while troubleshooting but without success. - Decision taken to POOH to change the BHA. Continue POOH to surface. - At surface the BHA was in good shape.
ZZ,807'	- Stopped drilling at depth ZZ807 ft and circulate one complete cycle. - Perform short wiper trip backream 10 stands and circulate complete cycle. - RIH to bottom on elevator, hole in good condition.

[Table-2](#) provides a summary of the geosteering activity conducted in the upper lateral drain using the automated downhole directional drilling and Geosteering workflow. This table includes data on the well path's trajectory, the geological formations encountered, and the adjustments made to optimize the drilling process. It also highlights key performance indicators, such as the accuracy of the steering decisions and the efficiency improvements achieved during the operation.

Table 2—Geosteering Activity

No	Depth (ft.MD)	Geosteering Decision
	XX761	Start drilling 6in lateral section Motor assy.
1	XY771	Based on the inversion, Formation bed dip is ~0.4° Up. Build and hold inclination at ~90.5° to maintain the well in target reservoir.
2	XZ956	The inversion is mapping a low-resistivity conductive layer ~ 6-7 ft TVD below the wellbore. Build and hold the inclination ~90.3° and 90.5° to maintain the well position ~ 15-17 ft TVD above the conductive layer.
3	YX181	The inversion is mapping a low-resistivity conductive layer ~11-12 ft TVD below the wellbore. Hold the inclination at ~ 90° to maintain the well within the target zone.
4	YY805	To drop and hold target inclination at 89° within the allowable DLS to place the well in target reservoir.
5	YZ070	To drop and hold target inclination at 88° within the allowable DLS to place the well in target reservoir.
6	YZ427	To drop and hold target inclination at 87.5° within the allowable DLS to place the well in target reservoir.
7	ZX105	To drop and hold target inclination at 88.5° within the allowable DLS to place the well in target reservoir.
8	ZY649	To drop and hold target inclination at 88° within the allowable DLS to place the well in target reservoir.

No	Depth (ft.MD)	Geosteering Decision
9	ZY775	To drop and hold target inclination at 89° within the allowable DLS to place the well in target reservoir.
10	ZY940	To drop and hold target inclination at 89.7° - 90° within the allowable DLS to place the well in target reservoir.
11	ZY434	To drop and hold target inclination at 89.5° - 89.7° within the allowable DLS to place the well in target reservoir.
12	ZY575	To drop and hold target inclination at 90° - 90.2° within the allowable DLS to place the well in target reservoir.
13	ZZ180	To hold target inclination at 89.5° within the allowable DLS to maintain the well in target reservoir.
14	ZZ495	To hold target inclination at 89.7° within the allowable DLS to maintain the well in target reservoir.
15	ZZ763	To hold target inclination at 89.4° within the allowable DLS to maintain the well in target reservoir.
	ZZ807	Well TD

The summary aims to offer a comprehensive overview of the geosteering efforts, showcasing both the challenges faced and the solutions implemented to navigate through complex subsurface conditions.

The automated downhole directional drilling and Geosteering workflow was fully executed in upper lateral drain with key information summarized in [Table-3](#).

Table 3—Upper Lateral Drain Performance Summary

Subject	Upper Lateral Drain
Target Zone	Carbonate Reservoir
Footage	6807 ft
Number of downlinks	74
On-bottom ROP	50.2 ft/h
Number of geosteering decisions	14

To compare the performance of upper and lower lateral drains, normalization is required for creating a reasonable and consistent comparison. The summary of comparison between upper and lower lateral drains are provided in this table:

Table 4—Performance Comparison

Subject	Upper Lateral Drain (Pilot)	Lower Lateral Drain	Average Data from Offset Wells
Footage Drilled (ft)	6807	3008	-
Downlinks (# per 1000ft)	10.9	29.7	16
Geosteering Decisions (# per 1000 ft)	2.2	5	5

Considering that the footage of upper lateral drain was 2 times longer than lower lateral drain, the number of downlink and geosteering decisions were normalized per 1000 ft. According to the trial results, the number of downlinks was reduced by 63%, and the number of geosteering decisions was reduced by 56%. Additionally, the trial results indicated reductions in both aspects compared to the average from offset wells. The comparison of geosteering decisions applied in both drains shown in the [Figure 6](#) below.

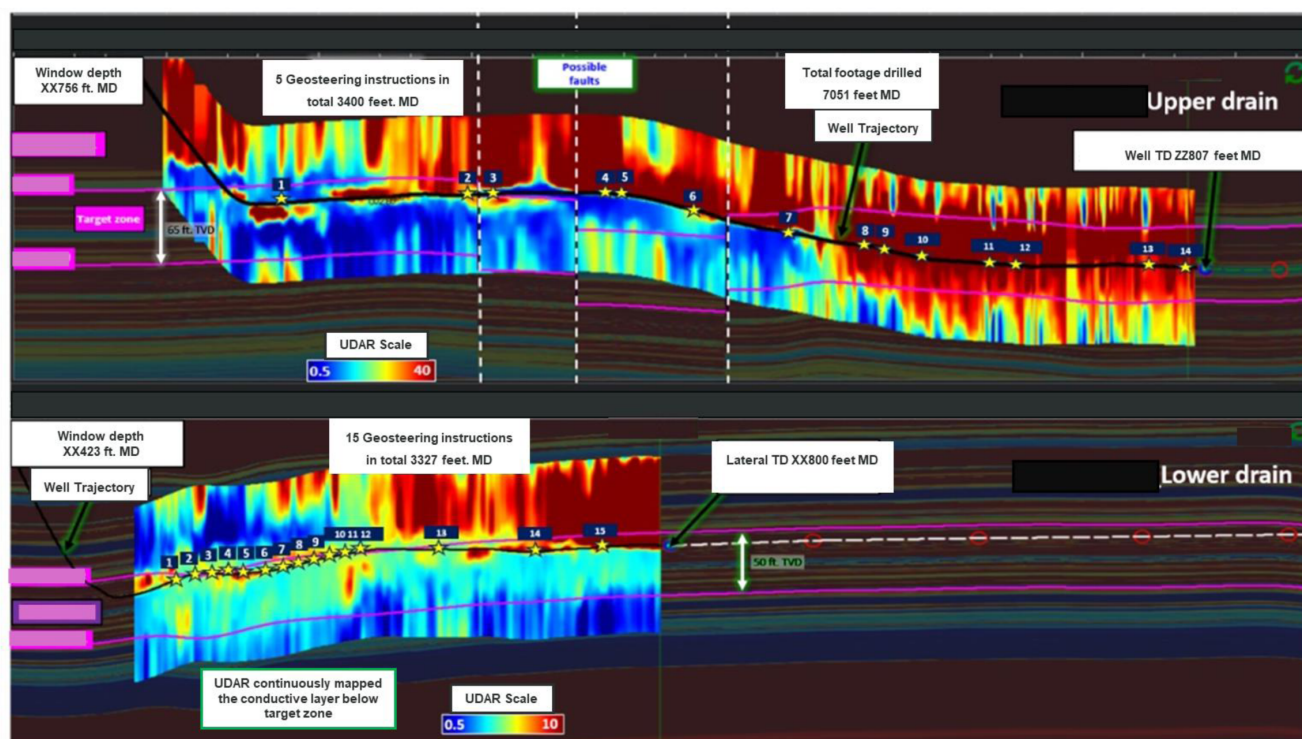


Figure 6—Geosteering Cross Section – Comparison between Upper and lower lateral

The on-bottom rate of penetration (ROP) in the upper lateral drain was calculated for a drilling interval of 5716 ft prior to controlling the ROP due to high torque (drilling with sea water) and experienced losses. The results indicated an increase of 19% in the on-bottom ROP compared to the lower lateral drain. To accurately assess the impact of downlink reduction and geosteering decisions on improving on-bottom ROP, a more detailed comparison is needed. This should involve scenarios where both drains face similar challenges in geosteering within the same target zone.

Table 5—OB ROP Performance Comparison

Subject	Upper Lateral Drain (Pilot)	Lower Lateral Drain
Footage Drilled (ft)	5716	3008
On-bottom ROP (ft/hr)	62.5	52.2

The pilot of the automated downhole directional drilling and Geosteering workflow successfully met the four key performance indicators (KPIs) outlined in the performance targets, as summarized in Table-6 below:

Table 6—Key Performance Indicators

Subject	Key Performance Indicator	Result
Downlinks	- 10%	- 63%
On-bottom ROP	+ 15%	+ 19%
# Geosteering decisions	- 17%	- 56%
Real Time Inclination and Azimuth Trend	80%	100%

Conclusion

The integration of automated downhole directional drilling and Geosteering workflow with directional drilling systems represents a transformative shift in modern drilling practices. These technologies significantly improve drilling precision and operational efficiency, enabling safer, more sustainable extraction of subsurface resources. By combining intelligent control algorithms with high-resolution sensor inputs, the system can guide the drill bit with exceptional accuracy, steering away from low-productivity zones and reducing the environmental footprint of drilling operations. At the core of this advancement is real-time data analysis, which empowers the system to dynamically adapt to changing downhole conditions. In highly variable or complex reservoirs, this capability is essential. The continuous stream of feedback from downhole sensors equips operators with actionable insights, allowing for immediate course corrections and well trajectory optimization—all while drilling is in progress. This integrated approach enhances drilling performance in several critical areas: it reduces drilling duration, boosts safety standards, improves reservoir contact, and optimizes resource utilization. By automating key decision-making processes, the system minimizes reliance on manual control, thereby reducing the potential for human error and increasing operational consistency. Greater accuracy also leads to more efficient use of materials and time, supporting both cost-effective operations and reduced environmental impact. Moreover, the adoption of these advanced systems unlocks new opportunities for drilling in geologically challenging or previously unreachable formations. The enhanced ability to collect and interpret subsurface data contributes to a more comprehensive geological understanding, ultimately improving site selection and long-term field development planning.

Acknowledgements

The authors would like to thank the management of ADNOC OFFSHORE, ADNOC Upstream and HALLIBURTON for the continuous support to implement the initiatives and for permission to publish this paper.